



Triathlon

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Performance Demands of Triathlon

This chapter describes the sport of professional triathlon, discussing the various factors of importance for success in the event, and how high-intensity interval training (HIIT) contributes to physical development in its three individual disciplines of swimming, cycling, and running.

Sport Description and Factors of Winning

Triathlon is considered by some to have its beginnings in the 1920s in France, in an annual sport called “Les trois sports.” However, the first modern-day triathlon occurred at Mission Bay, San Diego, California, on September 25, 1974, and has witnessed exponential growth to the present day, where about 2.5 million competitors now compete annually in the sport of triathlon in the United States alone. The sport of triathlon is an endurance race that involves successive swimming, cycling, and running (including transition time from one event to the next), and while it can be made up of a range of competition types, distances, and durations, the majority of races are broadly categorized into four distinct race distances, termed sprint, Olympic, half-Ironman, and Ironman triathlons (table 19.1).

The varying event distances in triathlon create specific technical, physiological, and nutritional considerations for athlete and practitioner alike. A range of factors, including terrain, environmental conditions, tactics, and strategy, influence the successful outcomes in these events. However, the primary predictor of performance in any triathlon event comes down to the physical capacities of the individual triathlete. These physiological attributes that determine success in the triathlon are similar to those in each individual sport of swimming (chapter 17), cycling (chapter 15), and running (chapter 14), and include those related to mode-specific qualities of aerobic power ($\dot{V}O_{2\max}$), economy, and fatigue resistance.

Due to the large variation in not only race distances (table 19.1) but also environment and topography,

Table 19.1 Primary Triathlon Distances for Swimming, Cycling, and Running

Event	Swim (m)	Bike (km)	Run (km)
Sprint	750	20	5
Olympic	1500	40	10
Half-Ironman	1900	90	21.1
Ironman	3800	180	42.2

there are equally subtle differences in the physiology type that can be competitive in triathlon. An example of this might be to contrast a winning sprint distance triathlete with a winning Ironman triathlete. The former athlete would be seen as generally younger, with a high $\dot{V}O_2\text{max}$, while the latter triathlete tends to be older, more experienced, and with higher levels of exercise economy and aerobic threshold levels (chapters 3 and 4). Rarely do we see athletes excelling in both distances in the spectrum at the same time of their life (i.e., winning both a sprint and Ironman triathlon in the same year). Often however we witness younger winners of sprint distance triathlons evolve to becoming successful at the Ironman distance as they age, such as 2008 Beijing Olympic champion and 2015 and 2016 Ironman champion, Jan Frodeno.

Additionally, triathlon is not without its skill requirements. Open-water swimming, although

similar to pool-based swimming's physical and technical requirements (see chapter 17, swimming), has its own set of unique requirements and challenges, including a marginally adjusted technique for open-water swimming, completed often with a wetsuit versus not (depending on temperature cut-off regulations), involving the navigation and battles among other competitors, including routing around the open course taking water current and conditions into account. Cycling varies considerably in its tactics due to different rule formats generally for sprint and Olympic-based events versus half-Ironman and Ironman distance races, the former two being draft-legal at the International Triathlon Union (ITU; sprint and Olympic triathlon governing body) level, while the latter two typically are not. The draft-legal ITU formats ultimately require a similar skill and tactic base as those in road cycling, while the latter involves the aerodynamics component of the indi-



a



b

(a) A younger 2008 Olympic Triathlon Champion Jan Frodeno of Germany and (b) the more mature 2015 and 2016 Ironman World Champion Jan Frodeno (right).

a. Adam Pretty/Getty Images; b. Warren Little/Getty Images for Ironman.

vidual time trial (see chapter 15, road cycling). Running can also be considered a technical sport (see related chapters), and the preceding swimming and cycling phases create unique demands on a triathlete's ability to run, affecting economy, kinematics (running form), and ultimately run performance (6-8, 12, 17). Finally, the clock does not stop between individual swim, bike, and run events, so triathletes must additionally be skilled at transitioning from one event to the next (8). While sport science has to date not been able to capture the impact of transitioning on overall performance, it stands to reason that, like with F1 car racing, the transition speed and smoothness could be impactful to overall performance, when you consider that finishing time at the end of a triathlon can be separated by less than a second.

While these aspects of the sport clearly require skill, practice, and experience, compared with sports such as tennis or basketball, the requirement for skill is comparatively low, making triathlon a sport primarily about the athlete's physiology. Therefore, the successful triathlon coach is one who understands the components that maximize triathlon physiology in the individual sports of swimming, cycling,

and running, while appropriately being able to blend training for the three sports (i.e., merging swimming, cycling, and running training, allowing suitable recovery). As shown in the swimming, cycling, and running chapters, HIIT is a key component for triathletes and is used at certain times to prepare for racing in each individual discipline.

Physical Performance Contributions to Winning

Contrasting many other physical capacity-based sports completed in shorter durations of less than 10 min (rowing, athletics, swimming, running), like road cycling, triathlons are competed over much longer durations (table 19.1). Consequently, prolonged endurance capacity is a key performance indicator (figure 19.1).

As mentioned, subtle differences do exist between the ITU (sprint and Olympic) versus Ironman-type events from the bike phase standpoint, where drafting in sprint and Olympic distance events creates more variation in degree of high-intensity outputs and race tactics. While certain tactics undoubtedly occur in half-Ironman and Ironman events (i.e., getting



Figure 19.1 The position of the triathlete (International Triathlon Union; ITU and Iron-distance) on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging this varies slightly between triathlon race and across exercise modes. Outside of critical psychological aspects, which are difficult to quantify, the pie chart shows the general relative importance of skills (15%), tactical awareness (5%), and physical capacities (80%) for triathlon success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

away on the bike), winning in these more prolonged races (3.5 to 8 h) arises more from well-paced sustained efforts (1). In contrast, winners in sprint (<1 h) and Olympic distance (<2 h) races often emerge from groups that form during the swim phase, and possibly extend their leads during the bike phase. Sometimes those who get in front during the swim phase are also highly motivated to push hard early during the bike phase to extend their leads over those who emerge later from the swim phase and may be less motivated to bridge the gap. Additionally, triathletes who have paced more appropriately from other groups can, more often than not, be seen coming from behind to catch runners who have expended substantial energy in their preceding swim and (mostly) cycling phases. Nevertheless, in these latter shorter events, the combined effect of drafting and tactics amid constantly changing terrain and course dynamics, alongside the variable energetic demands, results in critical periods of a race often dictating success (8). This implies that in the former short ITU formats, it is not always the athlete with the highest physical capacities who wins, with tactics and race execution also important.

Targets of Physical Performance in Triathlon

Undeniably, physical capacities specific to the triathlon event are important. The physiological factors of $\dot{V}O_2\text{max}$ and speed/power associated with first (VT_1) and second threshold (VT_2), introduced in chapter 3 and expanded on in chapter 4, play important roles to varying extents that are specific to the event distance. While these will vary, the general contribution of these parameters to race success is shown in figure 19.2. While not exclusive and highly individual, the general trends observed with sprint and Olympic distance triathletes are high levels of speed and power associated with $\dot{V}O_2\text{max}$ and VT_2 (8), while the older, more experienced long-distance triathletes tend to have higher speeds and power associated with VT_1 (11, 18).

In summary, to be competitive in triathlon, a triathlete requires physical characteristics that are specific to the race distance format, with varying degrees of speed/power associated with $\dot{V}O_2\text{max}$ and VT_1 and VT_2 (8, 19). While athlete age and genetic traits are large determinants of these physiological factors, they are developed and maximized with training (22), and especially in the case of the long-

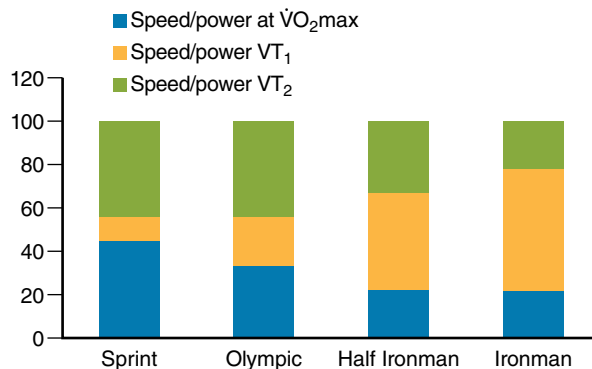


Figure 19.2 General physiological contributions to triathlon race success, including $\dot{V}O_2\text{max}$, first and second ventilatory thresholds (VT_1 , VT_2) for triathletes relative to the targeted event distance (table 19.1).

distance triathlete, integrated nutrient timing (33). Running is a critical component of the triathlon (8), and the neuromuscular characteristics needed to run fast and/or prolonged under fatigue are of key importance (8). HIIT is a key tool that can be used to target these specific characteristics.

Sprint

The sprint distance triathlon, performed at maximal intensity over about 45 min for males, and 55 min for females, requires a high-aerobic capacity in swimming, cycling, and running. As the top end of the intensity spectrum ($\dot{V}O_2\text{max}$ and VT_2 ; figure 19.2) are of likely key importance for this event (6), training that maximizes these capacities must be emphasized. As mentioned, successful sprint distance triathletes at the elite ITU level tend to be younger (18-30 yr), with very high sustainable capacities for short-duration swimming, cycling, and running.

Olympic

The Olympic distance triathlon, named for the format that first debuted at the Sydney 2000 Olympics, requires similar physiological characteristics as the sprint distance event (figure 19.2), and many athletes in the ITU often compete intermittently in both throughout a triathlon season. The double duration of the sprint event means that a greater emphasis is demanded on fatigue resistance and sustainable intensity. Due to the experience and time required to develop these properties, successful Olympic distance triathletes tend to be of a higher age (22-35 yr), although outliers are seen in both directions (21).

Half-Ironman

The half-Ironman distance event, from a physiological perspective considered to be ultra-endurance (16), requiring greater reliance on fat oxidation rates, carbohydrate sparing, and sustained endurance (figure 19.2), is completed in under 4 h for males and under 4.5 h for females. As with the former ITU events, specialists in half-Ironman tend to race both half-Ironman and Ironman distances, as they share similar characteristics. The half-Ironman event tends to be the first longer-distance event that an athlete graduating from ITU racing will attempt as he ages (general range of elite 27-38 years of age) and moves toward more prolonged racing. Thus, ITU athletes, with their high $\dot{V}O_{2\max}$ and VT_2 or critical power (CP) levels (chapter 4), are often immediately successful in the half-Ironman distance events (8). Present-day examples of half-Ironman winners include Olympians Alistair Brownlee (GBR) and Francisco Javier Gómez Noya (ESP).

Ironman

The Ironman event was born on February 18, 1978, off the beach of Waikiki, in Honolulu, Hawaii, when, to settle a bar bet, 15 competitors began a race to determine whether the fittest athlete was a long-distance swimmer, cyclist, or runner. The event combined three established single events in Hawaii, where the winner of all three in succession would be crowned the Ironman. It is a very long ultra-endurance event that is completed in around 8 h for top males and around 9 h for professional females. Like its half-distance counterpart, physiological attributes that enable Ironman success include high fat oxidation rates, $\dot{V}O_{2\max}$, and sustained endurance (11). These latter characteristics are reflected in high-aerobic threshold (VT_1) levels (19) (figure 19.2), and winners tend to be older and more-experienced triathletes (age range = 30-40).

Physical attributes of successful triathletes are dominated by aerobic capacities and thresholds described before (figure 19.2), as well as neuromuscular characteristics, specifically in running. Thus, HIIT sessions for triathletes are often used to develop aerobic metabolic and neuromuscular properties.

Key Weapons, Manipulations, and Surveillance Tools

For professional triathletes, most of their training hours (12-35 h/wk) are spent accumulating large

volumes of training at low exercise intensities (50%-80% of HRmax; 40%-70% $\dot{V}O_{2\max}$) (22). This relatively unstructured part of the training is important as it develops the aerobic foundation needed to support the development of $\dot{V}O_{2\max}$, economy, and thresholds, including high levels of fat oxidation rates (15, 16). As well as the large amounts of time training at low intensities, like cyclists (chapter 15), triathletes also complete specific pace work at medium, or mid-zone, intensities. These pieces, often performed within the middle of longer training sessions, are completed over a large range of durations that can range from 15 to 200 min at intensities of 80% to 90% HRmax or 70% to 80% $\dot{V}O_{2\max}$, depending on the athlete's age and experience and target event.

As with other endurance sports, while HIIT contributes considerably less time to a triathlete's training program, it is no less important. Work completed during HIIT is often either specific to race intensity or building on areas around race intensity (i.e., CP) to develop these components further. HIIT is vital for developing the adaptations of the oxidative (and nonoxidative) as well as the neuromuscular/musculoskeletal properties to enable fatigue resistance over prolonged durations (chapters 3, 4, and 5). The key weapons, or HIIT formats, including their manipulations and surveillance (monitoring) tools, will be described.

HIIT Weapons, Targets, and Manipulations

The most common HIIT weapons included in triathlon training programs, and specifically in swimming, cycling, and running, are primarily long intervals, short intervals, and sprint interval training, aiming for type 1 through 5 targets, with a metabolic emphasis, depending on the mode, individual triathlete characteristics, and stage of the program. Repeated sprint training, which incorporates very short duration efforts (3-7 s), is used occasionally in swimming across all athlete groups, and sometimes in biking and running by sprint distance triathletes. The breakdown of these formats is shown in figure 19.3.

Long Intervals

Long HIIT intervals usually take the form of 1 to 8 min work durations at intensities of 85% to 95% of HRmax, 85% to 100% of $p\dot{V}O_{2\max}$, generally with recovery intervals that range from 1 to 3 min

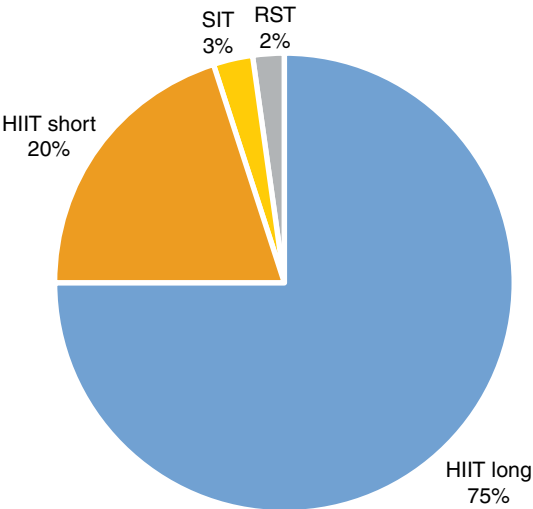


Figure 19.3 Percentage of the different HIIT formats (weapons) used throughout the annual season in professional triathlon.

(table 19.2). Total accumulated duration at these intensities during such interval sets is typically 12 to 30 min split between 1 and 5 sets. A specific long HIIT interval session (type 3 and 4 targets) for swimming might be 4×400 m in 5 min leaving on 7 min, 5×3.5 maximal maintainable power with 3 min passive (or extremely low intensity, <50 W) recovery for cycling, and 6×1000 m maximal maintainable run speed (~3 min) with 200 m recovery (passive or easy walk/jog) for running. Other good examples of long intervals specific for elite triathletes (modified appropriately) can be seen in the swimming (chapter 17), cycling (chapter 15), and running (chapter 14) chapters.

Short Intervals

Short HIIT intervals for triathletes usually take the form of 20 to 60 s work durations at intensities of

100% to 140% of $v/p\dot{V}O_{2max}$, generally with work:rest ratios between 1:0.5 and 1:1. Total accumulated work duration for these interval sets is between 10 and 20 min, split between 1 and 5 sets (table 19.3). An example of a short HIIT session (type 3 and 4 targets) for a triathlete would be 15×75 m completed at between 47 and 50 s per 75 m, leaving on 90 s (with a critical swim speed, CSS, of 1:13 per 100 m) for swimming, 2×(10×1 min at 110% of $p\dot{V}O_{2max}$ with 30 s recovery), 5 min RI for cycling, and 8 to 15×200 m on 32-34 s, 100 jog leaving on 60 s for running. Short intervals tend to be more emphasized for sprint and Olympic distance athletes over long-distance triathletes.

Sprint Interval Training

SIT is used infrequently by triathletes, but can be used in certain circumstances, and particularly when a triathlete possesses low absolute speed or power in any of the three individual disciplines. SIT is composed of maximal efforts between 10 and 30 s, with recovery periods of 2 to 10 min (table 19.4). As the goal of these intervals is to reach maximal values, the number of sprints is typically low (1-5), and time between sets relatively long (>30 min). For the start of sprint and Olympic distance triathlons, all-out swim speed is critical for reaching the first 300 m marker buoy before a bottleneck that consistently occurs after about the first 10 swimmers. Thus, SIT is mostly used in swim training (type 5 target).

Manipulations of Interval Training Variables

As with the track-and-field approach to HIIT session prescription (chapter 2), competition demands pertinent to sprint, Olympic, and long-distance triathlons are often used to develop specific swim, bike, and run HIIT sessions. Thus, nearly all of the 12 HIIT vari-

Table 19.2 HIIT Long-Interval Parameter Ranges for a Triathlete’s Swim, Bike, and Run Training

Mode	Work intensity	Work duration	Rest intensity	Rest duration	Rep number	Series	Between series recovery	Targets
Swim	102%-105% of CSS	200 m (~2:20-2:30)	0%-30% of CSS	90 s	4	2	100 m (2 min passive 30% CSS)	Type 4
Bike	95%-100% $p\dot{V}O_{2max}$	4 min	0%-30% $p\dot{V}O_{2max}$	3 min	5	0	Passive—40% $p\dot{V}O_{2max}$	Type 4
Run	95%-100% $v\dot{V}O_{2max}$	1000 m (3-3.5 min)	0%-30% $v\dot{V}O_{2max}$	200 m (90-120 s)	6	0	Passive—40% $v\dot{V}O_{2max}$	Type 4

CSS: critical swim speed; $p\dot{V}O_{2max}$: power associated with $\dot{V}O_{2max}$; $v\dot{V}O_{2max}$: velocity associated with $\dot{V}O_{2max}$

Table 19.3 HIIT Short Interval Parameter Ranges for a Triathlete's Swim, Bike, and Run Training

Mode	Work intensity	Work duration	Rest intensity	Rest duration	Rep number	Series	Between series recovery	Targets
Swim	108%-110% CSS	100 m (or 60-80 s)	0-30% of CSS	60 s	5	2	100 m (2 min passive 30% CSS)	Type 4
Bike	105- $\dot{V}O_{2\max}$	30 s	0%-30% $\dot{V}O_{2\max}$	30 s	10	3	5 min 40% $\dot{V}O_{2\max}$	Type 4
Run	103% $\dot{V}O_{2\max}$	30 s (or 200 m)	0%-30% $\dot{V}O_{2\max}$	30 s (or 100 m walk/jog)	10	2	5 min 40% $\dot{V}O_{2\max}$	Type 4

CSS: critical swim speed; $\dot{V}O_{2\max}$: power associated with $\dot{V}O_{2\max}$; $\dot{V}O_{2\max}$: velocity associated with $\dot{V}O_{2\max}$

Table 19.4 HIIT Sprint Interval Parameter Ranges for a Triathlete's Swim, Bike, and Run Training

Mode	Work intensity	Work duration	Rest intensity	Rest duration	Rep number	Series	Targets
Swim	All-out, around 114% of CSS	50 m (30-50 s)	passive	100 m or 2 min of passive recovery	6	0	Type 5 target
Bike*	All-out	15 s	passive	4 min	8	1	Type 5 target
Run*	All-out	20 s	passive	2 min	4	2	Type 5 target

* Infrequently used except potentially in particular instances of low absolute capacity (power or speed).

CSS: critical swim speed

ables (chapter 4) can be manipulated to tailor the acute physiological response (proxy adaptation).

The sprint and Olympic distance events in triathlon share some of the same issues as those outlined in the road cycling chapter (chapter 15). These shorter draft-legal formats add considerable variation to race demands, which can become unpredictable based on the dynamics and tactics of the race, and are affected by countless factors, including the outcome of the swim phase, the associated formation of varying numbers of groups, "attacks" or surges in pace to get away during the bike phase, the topography, turn number and turn type on criterium-style bike courses, as well as the effects of environment, including temperature, humidity, prevailing wind, and altitude. This array of factors, adding complexity to race outcomes, is somewhat less definitive in long-distance triathlon racing, although it can be still present to a degree. However, large surges in energy output, especially early on in a long-distance race, are often detrimental to overall performance outcomes, as the ability to sustain a high overall metabolic rate and movement speed throughout the event is reduced.

Notwithstanding these factors, the associated physiological parameters—CP, CSS, maximal lactate steady state (MLSS), functional threshold power/speed (FTP/S) (2), and ventilatory thresholds—are used to guide HIIT prescription. As with cycling, portable power meters are utilized to help guide and monitor cycling sessions, while run training sessions are often monitored using treadmill foot pods, outdoor global positioning system (GPS) or measured distances on a track, and follow the track-and-field approach to training prescription. Swimming sessions would be similar to those described in chapter 17 and are completed predominantly using measured times over set distances in the pool to determine a marker of swim specific critical power (chapter 4). These velocities can then be used to determine a CSS estimate (30), with intervals (i.e., time per 100 m) manipulated with these speeds in mind. A simple-to-use CSS estimate can be obtained from 200 and 400 m times using the following calculator (www.swimsmooth.com/training.html). The advantage to the aforementioned approaches (CSS and FTP/S) for triathletes is that strengths and weaknesses, typically evident in race

performance, can be manipulated within the individual mode (swim, bike, or run) to “close gaps” to required performance standards.

As outlined in chapter 4, the most defining factors to any HIIT weapon are the intensity and duration of the work and rest periods. Generally speaking, these should be in alignment with the demands of the event (table 19.1) and the identified performance gap of the individual triathlete (figure 1.5), so that the acute response of an HIIT session partially mimics the physiological demands that will be encountered in the specific event, and adaptation occurs appropriately. To take the two extreme ends of the triathlon distance continuum as an example, and speaking generally only, shorter duration and higher intensity bike and run intervals would be more frequently used for a sprint distance athlete, while more prolonged lower intensity efforts (very long and toward mid-zone) are applied with successful Ironman athletes (8, 15). Swim training programs tend to be similar across all triathlete distance focus groups.

Additionally, the number of intervals in a series, the series duration and time between each series of HIIT, will again depend on the specific competition demands (table 19.1) and the physiological/metabolic systems being targeted by the coach. As with cyclists (chapter 15) and runners (chapter 14), the use of terrain or hills, as well as cadence in the case of bike training, can be adjusted to engage certain metabolic and neuromuscular responses (23). The impact of cadence manipulation was described in the cycling chapter (chapter 15), while the use of hill training, sometimes referred to as functional resistance training (5), is described in road running (chapter 14). The neuromuscular demands of HIIT during swim training can be manipulated using tools that include paddles, bands, fins, and snorkels (chapter 17) (28).

Finally, as training sessions tend to be relatively long in duration for the triathlete, the decision of where to implement the HIIT session can also be important (start, middle, end, or various). A triathlete lacking in absolute speed or power may benefit by completing the HIIT session at the beginning of the session, while an athlete looking to improve fatigue resistance may benefit more by having the HIIT session toward the end of training. Typically, however, the placement of HIIT across a long-duration session tends to be random, relating more to the availability of suitable terrain (hills, flat section) or perception, feel, or motivation of the day.

Surveillance and Monitoring Tools

The main surveillance and monitoring tools in professional triathlon across all codes/events include primarily GPS-integrated wristwatches that capture aspects of external and internal training load (chapter 8). These devices combine running distance, time, pace, elevation change, and heart rate; power meters that record cycling power output, cadence, elevation change, and heart rate; and swim training data that includes lap distance, lap speed, and lap pace (and optional heart rate). These integrated wristwatch systems can stream data via Bluetooth and Wi-Fi seamlessly into associated analysis software systems (chapter 8). These tools can be used extensively for many applications that advise the coach and practitioner on a triathlete's discipline-specific and overall training load and training status.

External and Internal Training Load

Conveniently, the external training load for the triathlon disciplines of swimming, cycling, and running can be both prescribed and monitored using the aforementioned tools at hand. This prescription of course will vary based on the specific event discipline being trained for and the purpose of the session (i.e., aerobic development versus HIIT). For run and bike training, prolonged submaximal periods are typically prescribed using a combination of the RPE and heart rate method (i.e., run/ride at an easy/steady pace <130 bpm or VT_1 HR for 2 h), while low-intensity swim training is prescribed using pace or RPE (i.e., 500 m warm-up easy choice). The high-intensity and relatively short-duration HIIT intervals and the associated lag in HR response (see chapters 3, 8, and 9) prevent the use of HR for prescribing HIIT in any of the event disciplines. A more suitable approach is to use pace for running, using a measured track, foot pod/treadmill, or GPS-based device. Caution must be used when using such equipment, as GPS data can in our experience be variable due to a variety of environmental limitations, including cloud cover, terrain, industrialization, etc. (see chapter 8). As with road cycling (chapter 15), power output is easily measured (and viewed in real time by the athlete if desired) using power meters, with data seamlessly recorded (ANT+ or Bluetooth) and uploaded (Wi-Fi) to various software systems for immediate coach observation and analysis. Swim training mostly uses techniques described in chapter 17 where lap times are recorded across

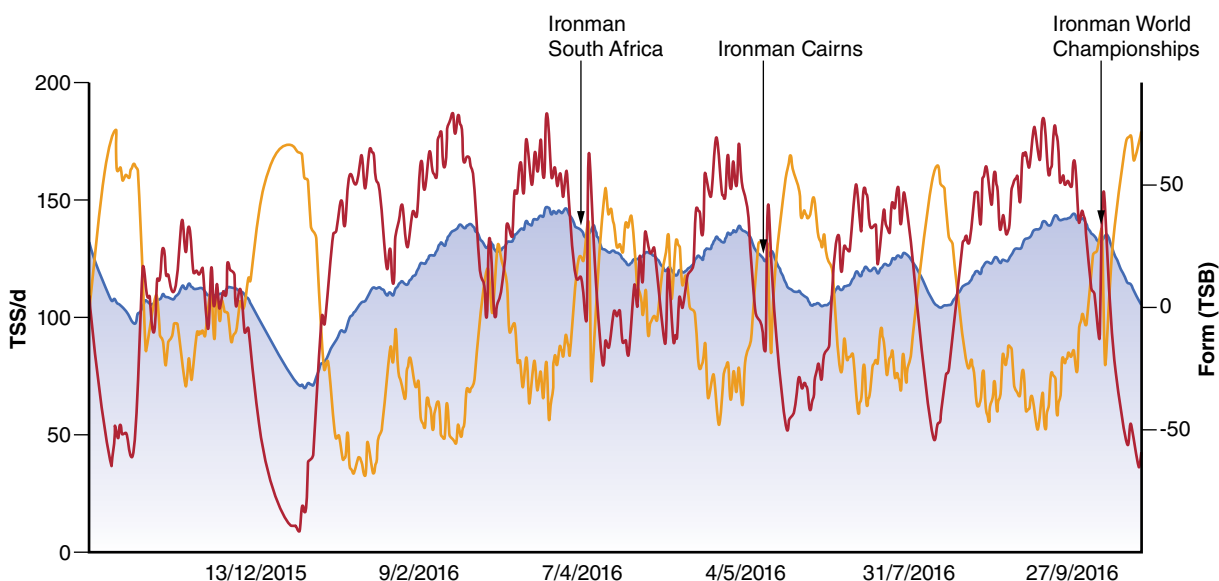


Figure 19.4 Training load monitored across one full season in an elite Ironman triathlete using the Training Stress Score available at TrainingPeaks.com. Blue shading represents the chronic training load (CTL; proxy fitness), the red line represents the acute training load (ATL; proxy training stress), while the orange line represents the training stress balance (TSB; proxy fatigue). As taught initially by Banister (4), this fitness minus fatigue equals form. Each peak in accumulated fitness (CTL) in this athlete represents the accumulation of training before a major event, finishing with the World Championships in Kona, Hawaii, in October.

known distances (either 25 m or 50 m pools). However, watch devices like Garmin are now able to record distance swum, pace, and stroke rate to give a general indication on swim training load to coaches when athletes are remote. For each of these triathlon disciplines, the training load can be estimated using various software systems that use Banister's classic fitness-fatigue modeling or training impulse (TRIMP) calculation scores in each respective discipline (3, 4, 21). Figure 19.4 shows the season-long training-load calculations for an elite Ironman triathlete.

As with other endurance-based sports, triathlon is rampant with the belief that more training is always better. Indeed, triathlon training volume, in terms of hours spent training each week, can range from as low as roughly 10 h in a starting junior sprint distance triathlete, up to reports of 40+ h in some Ironman distance triathletes. As a result, the risk of developing overtraining is high (chapter 7). The degree of training polarization, that is, the accumulation of time in an HR band associated with low intensity ($<VT_1$), mid-zone intensity (VT_1 - VT_2), and high intensity ($>VT_2$) work can also give insight and raise alarm bells with respect to the appropriateness of the response to the training (see following). In

our experience, we generally observe high degrees of training polarization (assessed by HR) with the accumulation of large amounts of time ($>85\%$) below VT_1 , $\sim 10\%$ in the mid-zone range, and $<5\%$ at high intensity.

Response to Training Load

As described in chapter 9, the body's response to the training load should also be monitored routinely to ensure the triathlete is progressing appropriately. Methods of response to training load can involve written training diary comments (13), direct face-to-face interaction (including phone/Internet conversations) with the triathlete discussing aspects of how training is feeling (feeling fast in the water, strong on the bike, light on the run as some examples), how he is sleeping, and his level of soreness. As described in chapter 9, morning resting heart rate variability (HRV; chapter 9) (24, 25) can also be used as a global assessment of the athlete's central nervous system response to training-induced stress, when assessed longitudinally against a history of morning measurements (25). A number of valid and reliable apps are now available for coaches to monitor this variable easily across groups of athletes (26).

Strategies for Structuring the Training Program Using HIIT

Implementing an HIIT program within the sport of triathlon is manageable, so long as races are scheduled to permit appropriate recovery between events. Nevertheless, there are a number of uncontrollable factors that must be managed to ensure appropriate effectiveness of the HIIT session.

Controllable and Uncontrollable Factors

Professional triathletes typically compete in 10 to 25 races each year. Such races are spread throughout the year and can depend on the personal location or home of the athlete. With key events (i.e., World Championships and Olympic Games) typically held in September or October each year, the majority of major event racing throughout the world typically is witnessed between the months of January and October.

Unlike a number of other sports (i.e., cycling, soccer, basketball, ice hockey), where competition frequency is high, the lower frequency of competition (if appropriately planned) enables sufficient recovery to enable the planned scheduling of HIIT. Smaller-level competitions are, however, used as a training stimulus leading into major events, to gain experience in more technical areas such as starts, pacing, tactics, transitions, and the competition environment experience.

Travel demands add another challenge to the training program that triathletes need to consider. Races are performed all over the world, including training camps in hot climates or at altitude (chapter 4), occasionally in difficult-to-reach places, including third-world countries that possess risk for infection from food and water sources that the triathlete may be unaccustomed to. Thus, proper planning for the location, along with recovery for and from travel (i.e., rest periods including before and after) as well as time for circadian adaptation to the time zone need to be considered (27). Here are additional strategies athletes should consider to lower the risk of illness, maintain comfort, and reduce symptoms of jet lag during long-haul travel to remote locations.

PREVENTING ILLNESS

- Avoid shaking hands or putting the hands to the eyes and nose.

- Wash hands with soap and water or an alcohol-based sanitizer (if soap and water are unavailable) before eating and after using the toilet.
- Wear a humidifying mask during flights.
- Drink plenty of fluids but avoid caffeine and alcohol.
- Be cautious of food on flights and in new destinations. Plan meals ahead, if possible.

MAINTAINING COMFORT DURING FLIGHTS

- For early flights when you will be more awake, book an aisle seat in advance so you can move around more frequently.
- For later flights during which you plan to sleep, book a window seat in advance.
- Wear compression socks and comfortable footwear during flights.
- Take regular short walks in the aisle and perform gentle stretches when possible to promote circulation.

REDUCING JET LAG BEFORE THE FLIGHT

- Be well rested before departure.
- Avoid HIIT 24 h before flying.
- Plan on-board and stop-over nutrition.
- Carry a drink bottle.

REDUCING JET LAG DURING THE FLIGHT

- Adjust to the new time zone immediately after takeoff, but still sleep when appropriate.
- If arriving early in the day, try to sleep during the second part of the flight. If arriving in the evening, sleep during the first part of the flight.
- During stopovers, sleep, walk, stretch, and shower, if time and space allow.

REDUCING JET LAG POST FLIGHT

- Perform easy aerobic training.
- Fall into your normal routine immediately, especially sleep, waking, meal patterns, and training times.
- Consume caffeine in the mornings only.
- Consider taking melatonin pills in the evening when traveling east.

Program Structure and Progression

There are a number of different approaches to triathlon training planning and preparation, and all have been used successfully at one time or another. These include classic periodization models, block periodization models, and numerous other formats (14).

The competition program for a triathlete is to some degree determined in accordance with her particular abilities/capacities and stage of development. As such, the triathlete's training program should work alongside the competition program, meaning that the training programs of triathletes are highly unique and specific to each individual athlete. Figure 19.4 shows how the training load and fitness gains in an elite Ironman triathlete fluctuate throughout the year based on the competition program, where each gain in predicted fitness corresponds with a major competition, finishing with the World Championship event at season's end. While each triathlete's program will be unique, some general observations can be made across the macro-, meso-, and microcycles.

Macrocycle

A triathlete's seasons are typically periodized around a small number of key racing objectives, requiring 2 to 8 periods of maximal performance. In ITU racing, for example, a top-level sprint or Olympic distance triathlete would typically aim to perform in 5 to 8 of the (current) nine World Triathlon Series events on offer throughout the year in order to compete in the World Series points system (cash/prize purse for most number of points in a season). In the World Triathlon Corporation (WTC), a brand that controls the majority of long-distance events, triathletes focusing on the Ironman World Championship in Kona may compete in from 2 to 5 Ironman races as well as a number of smaller-distance 70.3 events, and appropriately place emphasis on building toward those key performances, also in order to achieve entry in, and perform well in, the World Championship event (see figure 19.4). As mentioned, there are any other number of combinations of program structure that can be used that are formed appropriately to the individual athlete's goals and needs.

Mesocycle

Typical mesocycles for triathletes tend to range around 3 to 4 wk duration, with 2 to 3 loading weeks

and 1 recovery wk. However, a mesocycle could be as short as 10 d or as long as 12 wk, depending on the phase of the program and the length of time between the targeted events.

Microcycle

As with most athletes, a triathlete's microcycles tend to be weekly (7 d), although they can range from as short as 4 to as long as 10 days.

Periodized Nutrition

As outlined in chapters 4 and 7, periodized nutrition can play an important role in the life of many triathletes. Nutrition philosophies and practice range considerably across a wide spectrum, from chronic diets of very high carbohydrate (low fat) (10) to very low carbohydrate (high fat) (29), and every other combination in between. As outlined in chapter 4 (nutrition), triathletes often incorporate the practice of sleeping (20) and training low (9). For athletes using the ketogenic or low-carbohydrate high-fat (LCHF) diets to enhance power and speed at VT₁, some feel that the use of HIIT is vital to allow for the appropriate metabolic machinery to be maintained and stimulated (32). Interestingly, Webster et al. (31) recently showed a 2.8% enhanced 20 km cycling time trial performance in a 2 yr LCHF-adapted elite triathlete who supplemented 8 HIIT sessions over 3 wk with 60 g/h of carbohydrate during his HIIT sessions. Indeed, the use of carbohydrate supplementation for HIIT is, in our experience, a common practice for many elite fat-adapted triathletes.

Sample Training Programs

Tables 19.5 and 19.6 offer examples of an overall year training program, an example of a preparation phase and an example of a competition phase.

Acknowledgments

For any developing coach or scientist, it's foolish to fall into the type 2 trap (chapter 10) and think we know it all. We have tried our best not to, and are grateful to have been able to learn alongside some of the best coaches in the game, including Gordon Walker, Jon Brown, Pete Pfitzinger, Greg Fraine, and the late Laurent Vidal. Thank you, gents, for sharing your knowledge.

Table 19.5 Typical Training Week of an Olympic Distance Professional Triathlete During the Preparation Period: Hard and Easy Weeks

	Hard training week	Easy training week
Monday	a.m. 1: RUN 80 min LIT a.m.2: S&C p.m. 1: SWIM 3.5 k including HIIT 8×200 m, with 100 m passive recovery (type 3) p.m.2: BIKE 2 h LIT	a.m.1: RUN 50 min LIT a.m.2: S&C p.m.1: SWIM 3.5 k including HIIT (type 1) p.m.2: BIKE 1.5 h LIT
Tuesday	a.m.1: SWIM 3 k LIT a.m.2: RUN 40 min run LIT p.m.1: BIKE 60 min LIT p.m.2: RUN track HIIT 6×1000 m with 200 m recovery (type 3)	a.m.1: SWIM 3 k LIT a.m.2: RUN 20 min run LIT p.m.: RUN track HIIT 8×800 m with 200 m recovery (type 4)
Wednesday	a.m.1: SWIM 5 k LIT a.m.2: RUN 75 min LIT p.m.: BIKE 3.5 h LIT	a.m.1: SWIM 4.5 k LIT a.m.2: RUN 60 min LIT p.m.: BIKE 2.5 h LIT
Thursday	a.m.1: SWIM 5 k include 2×(8×50 MAX) on 90 s, 100 easy between series (type 4) a.m.2: RUN 60 min LIT p.m.: BIKE 2 h including 3×20 min MIT efforts (5 min recovery)	a.m.1: SWIM 5 k including 2×(10×100 on 1:45), 200 passive recovery between series HIIT (type 3) a.m.2: RUN 60 min LIT p.m.: BIKE 2 h including 4×10 min MIT efforts (4 min recovery)
Friday	a.m.1: SWIM 4.5 k LIT a.m.2: S&C p.m.1: RUN 60 min LIT p.m.2: BIKE 60 min LIT	a.m.1: SWIM 4 k LIT a.m.2: S&C p.m.1: RUN 60 min LIT p.m.2: BIKE 60 min LIT
Saturday	a.m.1: RUN 75 min including 6×5 min HIIT (5 min recovery) (type 3) a.m.2: BIKE 3 h LIT p.m.: RUN 30 min LIT	a.m.1: RUN 75 min including 4×5 min at 90% $\dot{V}O_{2\max}$ (5 min recovery) (type 4) a.m.2: BIKE 2.5 h LIT
Sunday	a.m.: BIKE 4 h LIT p.m.: RUN 1 h, 40 min LIT	a.m.: BIKE 3 h LIT p.m.: RUN 1 h, 15 min LIT

LIT: low-intensity training, blood lactate concentration <2.5 mmol/L, heart rate <80% HR_{max}; MIT: moderate-intensity training, blood lactate concentration 2.5-4.0 mmol/L, heart rate 80%-90% HR_{max}; HIIT: high-intensity interval training, blood lactate concentration >4.0 mmol/L, heart rate >90% HR_{max}; S&C: strength and conditioning

Table 19.6 Typical Training Week of an Olympic Distance Professional Triathlete During the Competition Period: Hard and Competition Weeks

	Hard training week	Competition week
Monday	a.m. 1: RUN 70 min LIT a.m. 2: S&C p.m. 1: SWIM 4 k including LIT p.m.2: BIKE 2 h LIT	a.m.: RUN 30 min run LIT p.m.1: SWIM 3.5 k including 4×200 (90 s recovery), 4×100 m (60 s recovery) (types 3 and 4) p.m.2: BIKE 2 h including 2×(10×30 s on/off) (type 3)
Tuesday	a.m. 1: SWIM 3 k LIT a.m. 2: RUN 40 min run LIT p.m.1: BIKE 60 min LIT p.m.2: RUN track 2×(10×400 m HIIT leaving on 1:45) (type 4)	a.m. 1: SWIM 3 k HIIT a.m. 2: RUN 20 min run LIT p.m.: RUN track 4×1000 m HIIT with 400 m recovery (type 4)
Wednesday	a.m. 1: SWIM 5 k LIT a.m. 2: RUN 75 min LIT including 2×20 min at MIT (10 min recovery) p.m.: BIKE 2.5 h LIT	a.m. 1: SWIM 3.5 k LIT a.m. 2: BIKE 2 h LIT p.m.: REST
Thursday	a.m. 1: SWIM 5 k including 10×200 m on 3:00 HIIT (type 3) a.m. 2: RUN 60 min LIT p.m.: BIKE 2 h, including weekly criterium racing (type 4)	a.m. 1: BIKE 90 min ride including 4×3 min HIIT with 7 min recovery (type 4) a.m. 2: RUN (off bike) 20 min including 2×3 min HIIT with 3 min recovery (type 4)
Friday	a.m. 1: SWIM 4.5 k LIT a.m. 2: S&C p.m.1: RUN 60 min LIT p.m.2: BIKE 60 min LIT	a.m. 1: SWIM 2.5 km include 6-7×100 m HIIT on 5 min (type 4) a.m. 2: BIKE 1.5 h LIT p.m.: RUN 30 min LIT
Saturday	a.m. 1: BIKE 3 h LIT a.m. 2: RUN (off bike) 75 min including 3×3 k HIIT, leaving on 15 min (type 4)	a.m. 1: SWIM 1000 m LIT a.m. 2: BIKE 60 min LIT p.m.1: RUN 15 min LIT
Sunday	a.m.: BIKE 3.5 h LIT p.m.: RUN 1 h, 30 min LIT	RACE DAY

LIT: low-intensity training, blood lactate concentration <2.5 mmol/L, heart rate <80% HRmax; MIT: moderate-intensity training, blood lactate concentration 2.5-4.0 mmol/L, heart rate 80%-90% HRmax; HIIT: high-intensity interval training, blood lactate concentration >4.0 mmol/L, heart rate >90% HRmax; S&C: strength and conditioning

CASE STUDY

Daniel Plews

While media coverage of triathlon predominantly focuses on stories of professional athletes, triathlon participation, which largely funds the sport, is predominantly made up of age-group (AG) triathletes. Here is my personal HIIT journey, the program I used to prepare for AG competition in the 2018 New Zealand Ironman triathlon.

AG triathlon competitions typically are broken into 5-year age categories (20-24, 25-29, 30-34, 35-39, etc.), which make the sport unique, permitting competition within competition as athletes strive to win their AG category. For AG athletes, who share full-time job and family commitments, training efficiency is everything. In this instance, it seemed even more important, as although I had completed four Ironman distance races previously, I had a new player in the mix, our little baby Bella, who would be 5 months old on race day. For context, the times of my previous Ironman races were 9 h 22 min (New Zealand, 2013), 9 h 08 min (New Zealand 2015), 9 h 12 min (Kona 2015), and 8 h 54 min (New Zealand, 2013).

Race day was March 3, 2018, and true preparation for IMNZ began on October 23, 2017 (19 wk). During the first 8 wk macrocycle, HIIT training was restricted to cycling only, with endurance, strength, and tempo training making up the swim, run, and other bike sessions. The first HIIT session of the week included long-interval HIIT (i.e., 3 (5×2 min at >5% critical power (CP)), 1 min recovery between reps, 4 min light spinning between sets; type 4 with type 3 short-interval HIIT (2 d later), composed of 20×60 s reps at >20% CP with 60 s recovery (light spinning). While to some in the triathlon community it might seem odd to start the training phase with such intense intervals, this is an approach I have used a number of times, where I progress from nonspecific (i.e., HIIT at powers well above IM pace) toward specific (lower intensity, longer duration) IM race pace reps. This sequencing continued for 8 wk, before the long-interval HIIT were progressed, eventually finishing with 8×4 min (>5% CP) with 2 min recovery (type 4). The follow-up short intervals were removed after 4 wk and replaced with intervals more specific to the half-Ironman distance (I was competing in half-Ironman distance races as part of my preparation). Although not technically HIIT sessions, these moderate-pace sessions consisted of longer-duration intervals (20-30 min) at an intensity ~15% below CP.

After this phase, which finished with a half-Ironman distance race, the cycle-based HIIT long intervals were reintroduced with increased durations. Sessions included 6×6 min at >5% CP and 4×8 min at >5% CP (type 4). Type 4 short-interval HIIT treadmill running immediately followed, consisting of 6 to 10×200 to 400 at 3:10 min/km pace (>15% CV). Sessions maintained throughout the entire training period included tempo runs (60 min running at around 85% HR_{max}) and strength sessions on the bike. These bike strength sessions, although not technically HIIT, allowed for periods at higher-power outputs and generally included 4 to 5×15 min hill ascents at 95% CP (7 min recovery).

The final 4 wk period was the most intensive period of training; all key sessions were touched on each week. To keep training as specific as possible, much time was spent performing very long intervals at moderate intensity (i.e., 4×30 min at 75% CP). HIIT was removed from the bike-training component, and the HIIT focus shifted to the run. Each week, one short (i.e., 6×200 m at 105% $\dot{V}O_{2\max}$, 400 m jog recovery; type 3) and one long (8×1 km with 200 m jog recovery at 100% $\dot{V}O_{2\max}$; type 4) interval session was completed. For me, this meant running the 1 km at 3:10/km (90 s recovery). My final "hit out" came 2 d prior to the race, an HIIT session I have done religiously before every race (I call it the Plews Primer) for as long as I can remember. This consists of 3×3 min at 102% CP with 4 min recovery (type 3). It always feels good to open the jets just before race day!

On race day, despite a small crash on the bike with 50 km to go, I couldn't have been happier with my result. My splits consisted of a 50 min swim, 4 h 47 min ride, and 2 h 52 min marathon run, resulting in an 8 h 35 min finish time, placing me 9th overall (including profes-

sionals) and first AG amateur overall by 15 min. This was also a personal best time over the Ironman distance by 20 min, and also broke the age-group course record by more than 10 min.

My actual weekly bike volume came out at only 261 km per week, which would surprise a lot of people. This is not a lot of riding relative to the time achieved. For the AG Ironman athlete, cycling is really the area that those of us who are time poor struggle with, and I certainly felt this with the demands of work and baby Bella. One of the reasons I believe I got away with such a low volume of riding was because I was very specific with the riding I was doing. HIIT was an integral part of achieving this outcome.

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